

Topic	Description
Module	Autonomous Mobile Robotics
Module Code	MECH-M-DUAL-3-AMR
Semester	WS 2025
Lecturer	Daniel T. McGuiness, Ph.D
ECTS	5
SWS	3
Lecture Type	ILV
Coursework Name	A Assignment
Work	Individual
Suggested Private Study	10 hours
Submission Format	Submission via SAKAI
Submission Deadline	09.02.2026 17:00
Late Submission	Not accepted
Resubmitting Opportunity	No re submission opportunity

No lecture time is exclusively devoted to the aforementioned assignment.

A portion of the mark for every assignment will be, where applicable, based on style. Style, in this context, refers to organisation, flow, sentence and paragraph structure, typographical accuracy, grammar, spelling, clarity of expression and use of correct IEEE style for citations and references. Students will find *The Elements of Style (3rd ed.)* (1979) by Strunk & White, published by Macmillan, useful with an alternative recommendation being *Economist Style Guide (12th ed.)* by Ann Wroe.

Submission Format and Requirements with L^AT_EX

Before submitting your work for assessment please follow the following requirements.

1. **The work must be done using L^AT_EX and no other format will be accepted.** You are allowed to use whatever software you wish to write (i.e., Overleaf, T_EXStudio, LyX, ...). The work must also conform to the **MCI Documentation guidelines** (which is in SAKAI). A template is provided to you [here](#) which was written by the author of this document.

If you are to use **mcidoc** template, set the `document-state` as `Report`.

To learn L^AT_EX, the author has tutorials which can be accessed [here](#).

Please treat this assignment as a great practice to learn L^AT_EX as you will need it when you are writing your thesis.

2. The report (and all its content) **must** be submitted as a `.pdf` along with all the necessary files (`.tex`, `.sty`) to compile it. Submit one (1) `.zip` file containing everything with the following name pattern.

`[STUDENT-ID]_[STUDENT-NAME]_[STUDENT-SURNAME].zip`

For example, a student named **Jason Brigham** with a student ID **2710704051** would send their submission as:

`2710704051_Jason_Brigham.zip`

and their submission would be in arranged as the following directory:

`2710704051_Jason_Brigham/`

```
├── tex/
│   ├── 2710704051_Jason_Brigham.tex, *.pdf, mcidoc.cls, custom.sty, ...
│   └── figures/
│       ├── .jpg, .png, ...
└── HW_Content/
    ├── src/, include/, .cpp, .bash, .py, ...
```

where `...` are any additional requirement (i.e., `.bib`, `.glo`) you may need for T_EX and the `HW_Content` is any content which is relevant to the Homework (i.e., if your HW is about python, this is where all the python scripts would be kept, or where your ROS 2 source files would be, or where your C++ codes would be for a project.)

3. The work must be submitted to SAKAI and **no link to personal repo will be accepted.**

To make sure your submission conform to given requirements, please test out zipping/un-zipping and running any code before submission. Any deviations from the requirements will incur penalties to the final grade.

Question	Maximum Point	Received Point
A Literature on Navigation Methods	100	
Sum	100	

[Q1] A Literature on Navigation Methods _____ 100

An important problem to tackle is the ability of a robot to find its path and map the area as it is navigating. This problem is further pronounced when the robot is sent to an environment it has no prior knowledge of.

For example, when we send a robot into a cave system, how can the robot know where it is? What objects will it use within its environment to re-calibrate itself and how can we make sure the relative coordinate of the robot is true to the global reference frame?

For this assignment you are to do scientific research on the topic of **map planning for mobile robots**. You will do a scientific analysis on what has been done and write a short article in $\text{\LaTeX}$ summarising it (i.e., Literature Review). This literature review should contain around 25 - 30 references containing recent research and development efforts on the topic of mapping, planning, and navigation. The cited papers must be published research.

Note: Your work will be analysed for novelty.

Guidelines and Requirements

1. Your citations must be done in IEEE format.
2. Your work must conform to the standards & requirements set by **Submission Format and Requirements with $\text{\LaTeX}$** .
3. For all citations you do, you need to include the page of the content source and highlight the cited part and put it into `HW_Content/src/document/XX.doc`, where `XX` denotes the citation number as shown in the homework.

Example: Let's say you have written a 5 page review paper encompassing about 25 citations. This means in your `HW_Content/src/document` folder, there should be `.pdf(s)`, with values ranging from `01.pdf` to `25.pdf`.

Second let's say you cite page 54 of a paper and this appears in your document as [5]. This means, the document should be cut so only page 54 would be left present, rename it to `05.pdf`, highlight the cited part, and then be placed into `HW_Content/src/document`.